

PHYSICS PRACTICAL SHEETS

Date: 20/01/26

PN CAMPUS

Class: BSc 2nd year

Experiment No.: 12

Roll No.: 68

Group: P2

Shift: Day

Sub.: Physics

Object of the Experiment (Block Letter)

Set: 4

OBJECT: TO DETERMINE SPECIFIC CHARGE FOR AN ELECTRON BY MAGNETRON METHOD.

APPARATUS REQUIRED:

- (i) Magnetron valve
- (ii) A standard long axial solenoid in which the diode can be mounted at the center.
- (iii) Power supply and cables

THEORY:

In the absence of a magnetic field, if V is the potential difference between the anode plate and the cathode, the kinetic energy gained by an electron on reaching the plate is given by,

$$eV = \frac{1}{2} mv^2$$

$$\text{or, } v = \sqrt{\frac{2eV}{m}} \quad \text{--- (I)}$$

where e is the electronic charge, v is the velocity of the electron and m is its mass.

When a critical field B_c is applied perpendicular to the direction of the velocity of the electron, the magnetic force experienced by it is equal to the centripetal force. That is,

$$B_c e v = \frac{mv^2}{R/2}$$

$$\text{or, } v = \frac{B_c e R a}{2m} \quad \text{--- (II)}$$

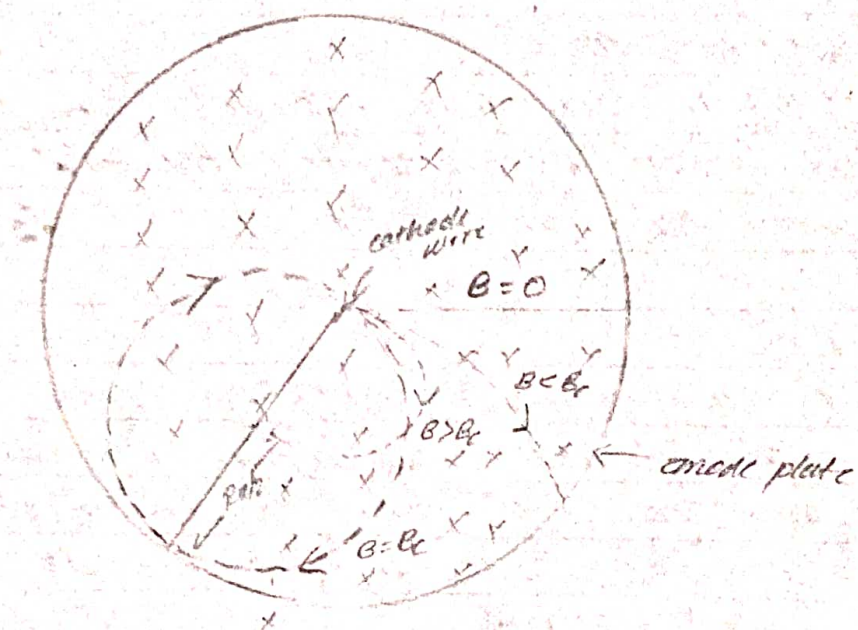


Fig: Cross section view of cylindrical magnetron tube

where, R_a is the radius of the cylindrical anode or the plate of the diode valve.

Comparing ① and ⑩,

$$\frac{B_c e R_a}{2m} = \sqrt{\frac{2eV}{m}}$$

Squaring both sides of this, we get

$$\frac{B_c^2 e^2 R_a^2}{4m^2} = \frac{2eV}{m}$$

$$\text{or } \frac{e}{m} = \frac{8V}{B_c^2 R_a^2}$$

The critical magnetic field B_c is produced by a solenoid of length L , the average radius R and the number of turns per meter n . If I_c is the current in the solenoid wire for critical field, then

$$B_c = \mu_0 n I_c \cos \theta$$

$$\text{where, } \mu_0 = 4\pi \times 10^{-7} \text{ Tm/A or (Wb/A.m)}$$

$$\text{and, } \cos \theta = \frac{L/2}{[R^2 + (L/2)^2]^{1/2}}$$

OBSERVATIONS:

The length of the solenoid, $L = 15.2 \text{ cm}$ and its radius, $R = 6.3 \text{ cm}$

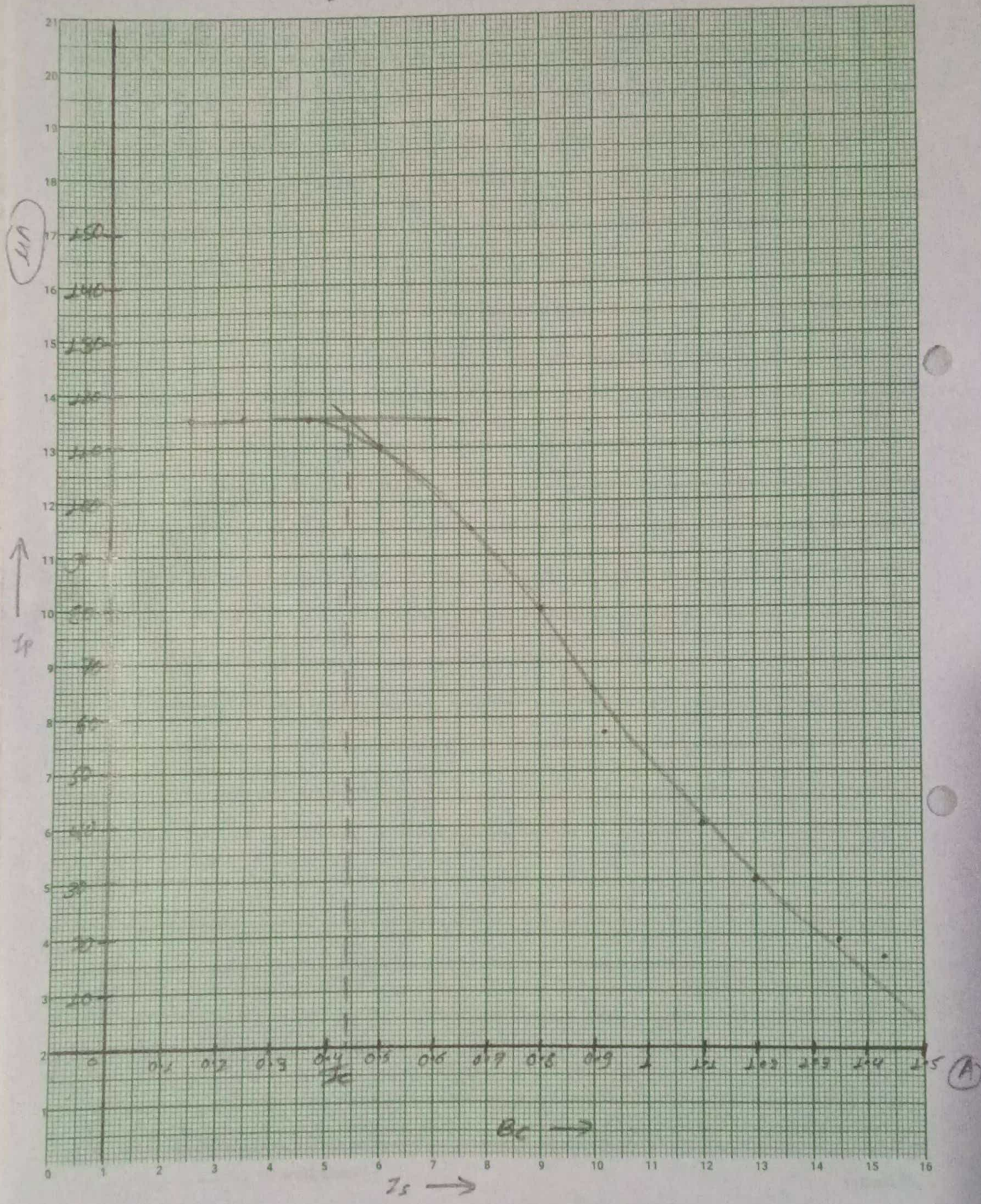
The number of turns per unit cm, $n = 11 \text{ turns/cm}$
 $= 1100 \text{ turns/m}$

The radius of the cylindrical anode,

$$R_a = 0.9 \text{ cm} = 0.009 \text{ m}$$

(I)

$$V_d = 0.4 \text{ volt}$$

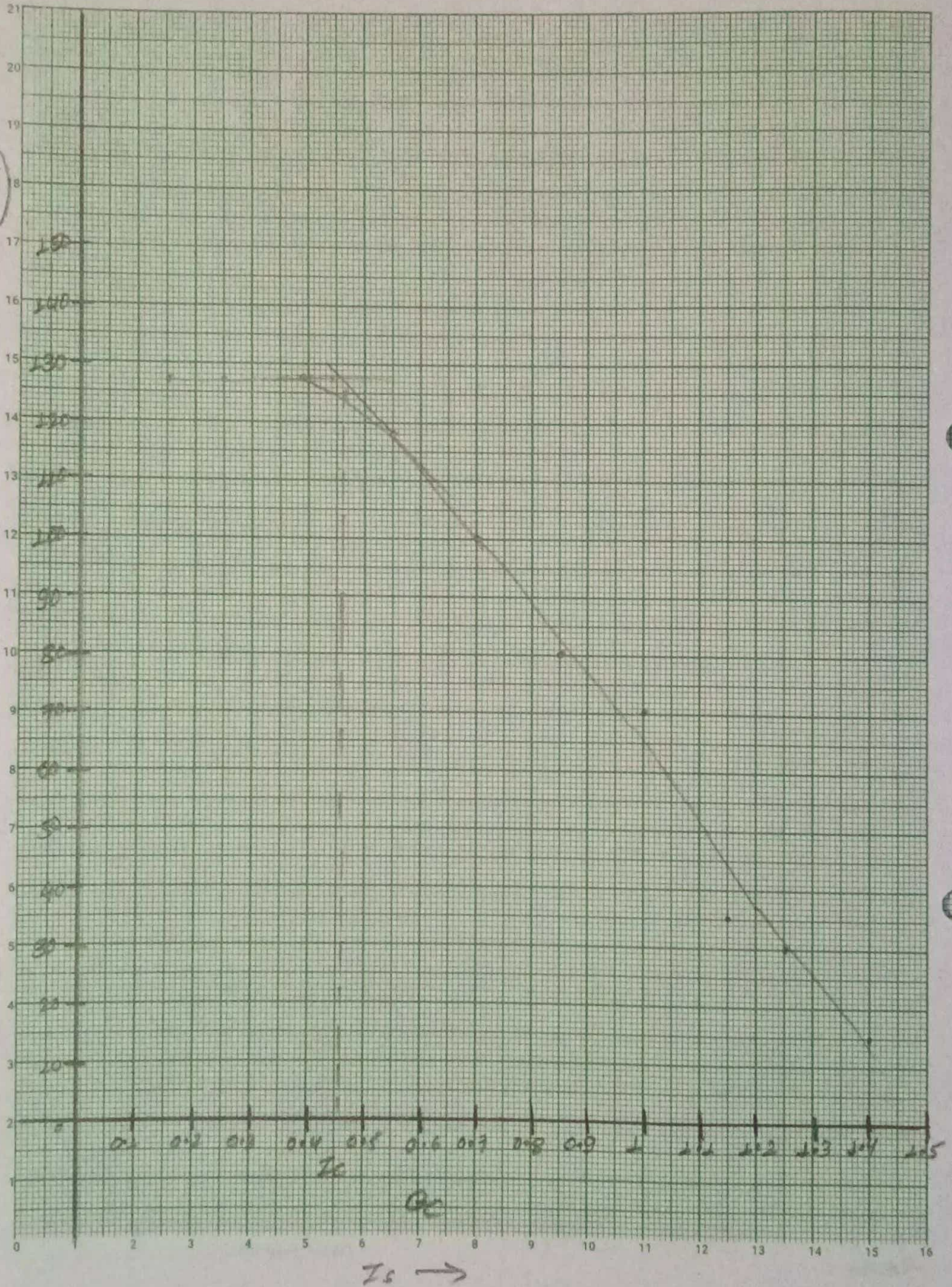


II

$$V_2 = 0.44 \text{ volts}$$

4.4

↑
R



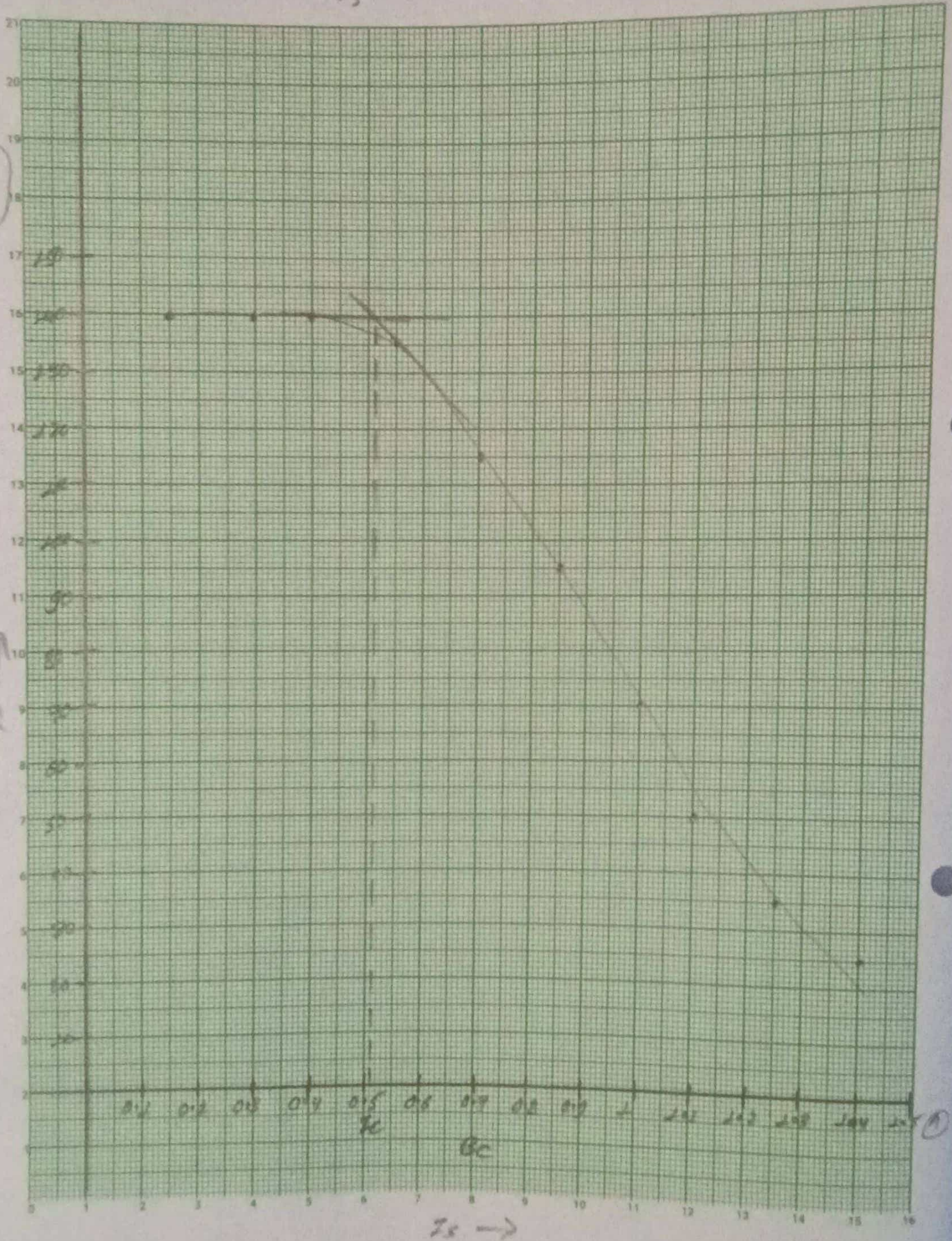
A

(10)

$V_g = 0.48$ volts

(11)

↑
11



$I_E \rightarrow$

(11)

OBSERVATION TABLE:

S.N.	Solenoid current in A for the plate potential (I_s)			Plate current in μA for the plate potential (I_p)		
	$V_1 = 0.4$	$V_2 = 0.44$	$V_3 = 0.48$	$V_1 = 0.4$	$V_2 = 0.44$	$V_3 = 0.48$
1.	0.15	0.25	0.25	115	127	140
2.	0.25	0.25	0.3	115	127	140
3.	0.37	0.4	0.4	115	127	140
4.	0.50	0.55	0.55	110	118	135
5.	0.67	0.7	0.7	95	100	115
6.	0.80	0.85	0.85	80	80	95
7.	0.92	1	1	57	70	70
8.	1.1	1.15	1.10	40	35	50
9.	1.2	1.25	1.25	30	30	35
10.	1.35	1.4	1.4	19	20	25
11.	1.43	—	—	16	—	—

CALCULATION:

The magnetic permeability, $\mu_0 = 4\pi \times 10^{-7} \text{ Tm/A or Wb/(A.m)}$

$$\text{and, } \cos \theta = \frac{L/2}{\sqrt{R^2 + (L/2)^2}} = \frac{15.2/2}{\sqrt{6.3^2 + (15.2/2)^2}} = 0.77$$

From the graph, the critical solenoid,

$$\textcircled{i} \quad I_{c_1} = 0.44 \text{ A}$$

$$\textcircled{ii} \quad I_{c_2} = 0.46 \text{ A}$$

$$\textcircled{iii} \quad I_{c_3} = 0.51 \text{ A}$$

The critical magnetic field, $B_c = \mu_0 n I_c \cos \theta$

$$\textcircled{i} \quad B_{c_1} = \mu_0 n I_{c_1} \cos \theta = 4\pi \times 10^{-7} \times 1180 \times 0.44 \times 0.77 = 4.683 \times 10^{-4}$$

$$\textcircled{ii} B_{c2} = \mu_0 n I_{c2} \cos \theta = 4\pi \times 10^{-7} \times 1100 \times 0.46 \times 0.77 \\ = 4.896 \times 10^{-4}$$

$$\textcircled{iii} B_{c3} = \mu_0 n I_{c3} \cos \theta = 4\pi \times 10^{-7} \times 1100 \times 0.51 \times 0.77 \\ = 5.428 \times 10^{-4}$$

and the specific charge, $\frac{e}{m} = \frac{8V}{B_c^2 R_0^2}$

$$\textcircled{i} e/m = [(8 \times 0.4) / (4.683 \times 10^{-4})^2 \times (0.009)^2] = 1.80 \times 10^{11} \text{ C/kg}$$

$$\textcircled{ii} e/m = [(8 \times 0.44) / (4.896 \times 10^{-4})^2 \times (0.009)^2] = 1.81 \times 10^{11} \text{ C/kg}$$

$$\textcircled{iii} e/m = [(8 \times 0.48) / (5.428 \times 10^{-4})^2 \times (0.009)^2] = 1.6 \times 10^{11} \text{ C/kg}$$

$$\Rightarrow \text{Mean } e/m = 1.736 \times 10^{11} \text{ C/kg}$$

RESULT:

The specific charge of an electron was found to be $1.736 \times 10^{11} \text{ C/kg}$ - using magnetron method.

ERROR:

$$\text{error \%} = [(1.78 - 1.736) / 1.78] \times 100 \% = 2.47 \%$$

PREF. CONCLUSION:

The specific charge of an electron was found to be $1.736 \times 10^{11} \text{ C/kg}$ with 2.47% error.

PRECAUTIONS:

- ① The valve should be kept at the center of the solenoid.
- ② The voltage applied at the plate of the valve should not exceed the maximum permissible rating of the valve.
- ③ The current in the solenoid at its critical field value should not be allowed to flow for a long time as it would heat up the wires.